Meesala Tarunsai

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2025	8.36
Sri Chaitanya Junior college	Class XII (Board Of intermediate Education)	2019	93.60%
Bhaskar High School	Class X (Board of secondary education)	2017	9.50%

POSITIONS OF RESPONSIBILITY

Teaching Assistant, Department of Mechanical and Aerospace Engineering, IIT Hyderabad: conducted **Statics and Dynamics laboratory experiments** for **B.Tech students**, guiding them through setup, data collection and result analysis

SKILLS

FEA / Simulation: Abaqus, Ansys Workbench, Motor-CAD

Experimental / DAQ: NI DAQ, LabVIEW, Accelerometers, Microphones,M+P analyzer

CAD and Programming: Autodesk Fusion 360, MATLAB

RELEVANT COURSES

Vibrations Engineering | Engineering Noise Control | Advanced Solid Mechanics | Finite Element Method

PROJECTS

NVH Investigation of a BLDC Motor: Structure-Borne Noise Root Cause and Sound Quality

M.Tech Thesis | IIT Hyderabad | Guide: Prof. Venkatesham Balide [2026MAR-Present]

- Measured **sound and vibration** using **microphones and accelerometers** with **NI DAQ and LabVIEW**, and identified **matched FFT peaks** indicating **structure-borne noise**
- Found the **root cause** by correlating **Motor-CAD/Maxwell electromagnetic forces** with **Experimental Modal Analysis** and **Ansys modal/harmonic analysis** to separate **force excitation** from **resonance**
- **Synthesized the motor sound** and evaluated **sound quality metrics** (**loudness, sharpness, roughness**)

Tools: LabVIEW, NI DAQ, Motor-CAD, Ansys Workbench

Concepts: NVH, FFT, Modal Analysis, Electromagnetic Forces, Resonance, Sound Quality

Noise Source Ranking and Reduction of a Hair Dryer

Course Project | IIT Hyderabad | Guide: Prof. Venkatesham Balide [2026 JAN-2026 APRIL]

- Classified **structure-borne vs. air-borne sources** using **vibration-acoustic spectra** and an **inlet-blocking test** (**79 to 65 dB**)
- Achieved a **6 dB noise reduction** (**79 to 73 dB**) using a **paper-straw flow straightener** at the exhaust outlet

Tools: MATLAB, Microphone, Accelerometer

Concepts: Blade-Pass Frequency, SPL, 1/3 Octave Analysis, Noise Reduction

Stress Analysis of Healthy and Keratoconus Corneas

Course Project | IIT Hyderabad | Guide: Prof. Vishwanath [DEC 2025]

- Built **3D nonlinear FE models** using **Mooney-Rivlin hyperelasticity** and **C3D8 elements**
- Compared **apex displacement and stress** at **10 mmHg IOP**, showing higher deformation and a **high-stress ring** in keratoconus

Tools: Abaqus, MATLAB

Concepts: FEA, Mesh Selection

Magnesium-Alloy Cylinder Head Analysis for a Dual Fuel Engine

B.Tech Project | Andhra University | Guide: Prof. K. Ramji [2022 DEC-2023 MAY]

- Performed **static structural and steady-state thermal analysis** in **Ansys Workbench** on a **4-cylinder CI engine cylinder head**, comparing **A-356 aluminium with WE43 and WE54 creep-resistant magnesium alloys** for **weight reduction**
- Evaluated **total deformation, von-Mises stress, temperature distribution and heat flux** to assess the **structural and thermal performance** of each material

Tools: Ansys Workbench

Concepts: FEA, Thermal Stress, Material Selection, Lightweighting

EXPERIENCE

Vocational Trainee, Central Machine Shop (CMS) | Rashtriya Ispat Nigam Limited (Vizag Steel), Visakhapatnam

Mentor: Sri B. Venkateswara Rao (AGM, CMS) [13 Feb 2023 - 11 Mar 2023]

- Studied and worked on **dynamic balancing of rotating elements**, understanding **unbalance, vibration and balancing procedures** used in plant machinery
- Observed **machine shop operations** and prepared an **internship project report** on the topic

Concepts: Rotor Dynamics, Unbalance, Dynamic Balancing, Vibration

EXTRACURRICULAR

Sports : Cricket , badminton, chess